



College of Engineering, Construction and Living Sciences
Bachelor of Information Technology
ID721001: Mobile Application Development
Level 7, Credits 15
Project

Assessment Overview

In this **individual** assessment, you will develop two mobile applications using **React Native** and **Expo**.

Learning Outcomes

At the successful completion of this course, learners will be able to:

1. Implement and publish complete, non-trivial, industry-standard mobile applications following sound architectural and code-quality standards.
2. Identify relevant use cases for a mobile computing scenario and incorporate them into an effective user experience design.
3. Follow industry standard software engineering practice in the design of mobile applications.

Assessments

Assessment	Weighting	Due Date	Learning Outcomes
Practical	20%	Sunday 9th November at 11.59 PM	1, 2
Project	80%	Sunday 9th November at 11.59 PM	1, 2, 3

Conditions of Assessment

You will complete this assessment mostly during your learner-managed time. However, there will be time during class to discuss the requirements and your progress on this assessment. This assessment will need to be completed by **Sunday 9th November at 11.59 PM**.

Pass Criteria

This assessment is criterion-referenced (CRA) with a cumulative pass mark of **50%** over all assessments in **ID721001: Mobile Application Development**.

Submission

You **must** submit all files via **GitHub Classroom**. Here is the URL to the repository you will use for your submission – https://classroom.github.com/a/Mq0MeQ_x. If you do not have not one, create a **.gitignore** and add the ignored files in this resource - <https://raw.githubusercontent.com/github/gitignore/main/Node.gitignore>.

Create a branch called **project**. The latest files in the **project** branch will be used to mark against the marking rubric. Please test before you submit. Partial marks **will not** be given for incomplete functionality. Late submissions will incur a **10% penalty per day**, rolling over at **12.00 AM**.

Authenticity

All parts of your submitted assessment **must** be completely your work. The use of **AI tools** is not permitted in this assessment. You need to demonstrate to the course lecturer that you can meet the learning outcomes for this assessment. You **must** complete the <https://moodle.op.ac.nz/mod/page/view.php?id=1178172> form and put this in the root of your repository. Failure to do this may result in a mark of **zero** for this assessment.

Policy on Submissions, Extensions, Resubmissions and Resits

The school's process concerning submissions, extensions, resubmissions and resits complies with **Otago Polytechnic** policies. Learners can view policies on the **Otago Polytechnic** website located at <https://www.op.ac.nz/about-us/governance-and-management/policies>.

Extensions

Familiarise yourself with the assessment due date. Extensions will **only** be granted if you are unable to complete the assessment by the due date because of **unforeseen circumstances outside your control**. The length of the extension granted will depend on the circumstances and **must** be negotiated with the course lecturer before the assessment due date. A medical certificate or support letter may be needed. Extensions will not be granted on the due date and for poor time management or pressure of other assessments.

Resits

Resits and reassessments are not applicable in **ID721001: Mobile Application Development**.

Instructions

Note: If you wish to develop different applications than those specified below, you **must** get approval from the course lecturer first.

Functionality - Learning Outcomes 1, 2 (20%)

- Application v1 with the following functionality:
 - Data is fetched from a local **JSON** file. You have been provided with a sample file called **application-development-data.json**.
 - App icon and splash screen.
 - Styled using **Tailwind CSS**.
 - **Recipe Details Screen Features:**
 - * Display the recipe's name, cuisine, image, ingredients, difficulty, prep time, cook time, servings, tags, nutrition, average rating and reviews.
 - * Display a heart icon in the top-right corner. Tapping it adds or removes the recipe and saves the changes.
 - * Display a plus icon next to the heart icon. Tapping it adds or removes the recipe's ingredients and saves the changes, ensuring no duplicate ingredients are stored.
 - **Daily Specials Screen Features:**
 - * Display a list of five randomly selected recipes with difficulty level Easy or Medium.
 - * Each item should show the recipe's name, cuisine, image and average rating.
 - * Tapping an item should navigate to the **Recipe Details Screen**.
 - **All Recipes Screen Features:**
 - * Display a list of all recipes.
 - * Each item should show the recipe's name, image, difficulty level, total time and average rating.
 - * Tapping an item should navigate to the **Recipe Details Screen**.
 - **Favourites Screen Features:**
 - * Display a list of saved favourite recipes.
 - * Each item should show the recipe's name, image and average rating.
 - * Tapping an item should navigate to the **Recipe Details Screen**.
 - **Ingredients Screen Features:**
 - * Display a list of saved ingredients grouped by category.
 - * Each item should display the ingredient name, quantity and category.
 - * Display a trash icon. Tapping it removes the ingredient and saves the changes.
- Application v2, which builds on v1 with the following functionality:
 - Select any four **Medium Features** from the list below:
 - * Filter recipes by cuisine, tags, difficulty level and ingredient categories.
 - * Search recipes by name, ingredient name or tag.
 - * Sort recipes by prep time, cook time, total time, difficulty level or average rating.
 - * Dark mode support with a theme toggle.
 - * Delete all saved favourite recipes or ingredients.
 - * Display recipes by times cooked or last cooked.
 - * Display a list of the last five recipes viewed.
 - Select any four **Hard Features** from the list below:
 - * Save and edit notes on recipes.
 - * Add ratings and write reviews for recipes.
 - * View combined ratings and user reviews.

- * Manage user profiles with preferences and dietary needs.
- * Add new recipes.
- * Update and delete existing recipes.
- * Meal planner to schedule recipes by day.
- Choose two features of your choice. These features should be different from the ones listed above.
- Create an **APK** for both v1 and v2.

Code Quality and Best Practices - Learning Outcomes 1, 3 (35%)

- Write clean, readable, and maintainable code.
- Use modular, reusable components and avoid code duplication.
- Keep code simple and easy to understand.
- Ensure the applications are responsive across different screen sizes and orientations.
- Optimise performance by managing resources efficiently.

Version Control - Learning Outcome 3 (5%)

- Regular commits are made throughout development, demonstrating meaningful progress.
- Commit messages are clear and descriptive.
- The repository shows a clean and logical commit history that reflects the development process of the applications.

Testing - Learning Outcome 3 (10%)

- Test the applications with three different users for each version (six users total), observing how they navigate and use features.
- Collect written feedback on usability issues, confusing UI and bugs.
- Write a brief report summarising the user testing, including:
 - Tester information (anonymised).
 - Tasks performed.
 - Positive feedback received.
 - Areas for improvement identified.
 - Bugs or errors found.
 - Suggestions on how feedback could be used to improve the applications (no requirement to implement).

Reflection - Learning Outcomes 1, 3 (5%)

- Write a short reflection (300-500 words) on your development process and learning throughout the project, including:
 - Challenges you faced and how you overcame them.
 - What you would do differently in future projects.
 - New skills, tools or practices you learned during the project.

Presentation - Learning Outcome 2 (5%)

- Record a 5-10 minute screen recording of your applications.
- Demonstrate the v1 and v2 features.
- Make sure the video is clear and easy to follow.
- Submit a link to the presentation in your repository's **README.md** file.

Marking Rubric

Functionality - Learning Outcomes 1, 2 (20%)

Criteria	Excellent (A)	Good (B)	Satisfactory (C)	Not Yet Achieved (D)
Application v1 - Data Fetching, Icon and Styling (2%)	Data fetched correctly from local JSON file. App icon and splash screen professionally designed and implemented. Fully responsive UI using Tailwind CSS across all screens.	Data fetched correctly. App icon and splash screen present. Responsive UI with minor styling issues.	Data fetched but may have issues. Basic icon and splash screen present. Partially responsive UI with styling gaps.	Data fetching fails or incomplete. Icon/splash screen missing or non-functional. Poor or no responsive design.
Application v1 - Recipe Details Screen (3%)	All required fields displayed correctly (name, cuisine, image, ingredients, difficulty, prep time, cook time, servings, tags, nutrition, average rating, reviews). Heart icon toggles favourites perfectly. Plus icon manages ingredients with no duplicates. Data persists correctly.	Most required fields displayed. Heart icon functional with minor issues. Plus icon works but may have duplication issues. Data mostly persists.	Some required fields displayed. Heart or plus icon partially functional. Basic data persistence with gaps.	Multiple required fields missing. Icons non-functional or missing. No data persistence.
Application v1 - Daily Specials Screen (2%)	Five random recipes displayed with Easy or Medium difficulty only. All required fields shown (name, cuisine, image, average rating). Navigation to Recipe Details works perfectly. Randomization works on each visit.	Five recipes displayed with mostly correct difficulty filtering. Most required fields shown. Navigation functional with minor issues.	Recipes displayed but difficulty filtering incomplete or random selection not working. Some required fields shown. Basic navigation present.	Fewer than five recipes or no difficulty filtering. Required fields missing. Navigation non-functional.
Application v1 - All Recipes Screen (2%)	All recipes displayed in list format. All required fields shown (name, image, difficulty, total time, average rating). Navigation to Recipe Details works perfectly. Performance optimized for large lists.	All recipes displayed. Most required fields shown. Navigation functional with minor issues. Adequate performance.	Recipes displayed with some required fields. Basic navigation present. Performance issues with scrolling.	Recipe list incomplete or non-functional. Required fields missing. Navigation broken.

Criteria	Excellent (A)	Good (B)	Satisfactory (C)	Not Yet Achieved (D)
Application v1 - Favourites Screen (2%)	All saved favourites displayed correctly. All required fields shown (name, image, average rating). Navigation to Recipe Details works perfectly. Updates reflect immediately when favourites change.	Favourites displayed with most required fields. Navigation functional. Updates mostly reflect in real-time.	Basic favourites list present. Some required fields shown. Navigation works but updates may be delayed.	Favourites feature non-functional or missing. Required fields missing. No real-time updates.
Application v1 - Ingredients Screen (2%)	Ingredients grouped correctly by category. All required fields shown (name, quantity, category). Trash icon removes ingredients and persists changes perfectly. Groups update dynamically.	Ingredients displayed and grouped with minor issues. Most required fields shown. Trash icon functional with some persistence issues.	Basic ingredients list with incomplete grouping. Some required fields shown. Trash icon partially functional.	Ingredients not grouped or missing. Required fields missing. Delete functionality non-functional.
Application v2 - Medium Features (3%)	Four medium features fully implemented and functional. Features are well-integrated, polished and enhance user experience significantly.	3-4 medium features implemented with minor issues. Features mostly integrated and add value.	2-3 medium features present with functionality gaps. Basic integration.	Fewer than 2 medium features or non-functional. Poor integration or no user value.
Application v2 - Hard Features (3%)	Four hard features fully implemented and functional. Features are complex, well-integrated and significantly enhance app functionality.	3-4 hard features implemented with minor issues. Features add substantial value with some limitations.	2-3 hard features present with functionality gaps. Basic implementation.	Fewer than 2 hard features or non-functional. Poor implementation or incomplete functionality.
Application v2 - Custom Features (1%)	Two custom features fully implemented and functional. Features are creative, original and enhance the application.	1-2 custom features implemented with minor issues. Features add some value.	1 custom feature present with functionality gaps. Basic implementation.	Custom features missing or non-functional. No added value.
APK Creation (2%)	Both v1 and v2 APKs successfully created, installable and fully functional. Professional build configuration.	Both APKs created with minor issues. Mostly functional after installation.	Both APKs created but with functionality or installation issues.	APKs missing, won't install, or only one version created.

Code Quality and Best Practices - Learning Outcomes 1, 3 (35%)

Criteria	Excellent (A)	Good (B)	Satisfactory (C)	Not Yet Achieved (D)
Code Readability and Maintainability (9%)	Code is exceptionally clean and readable with consistent formatting. Meaningful variable and function names. Well-organised file and component structure. Comprehensive comments where needed.	Code is clean and readable with good naming conventions. Organised structure with some helpful comments. Minor inconsistencies.	Code is mostly readable. Some unclear naming or inconsistent formatting. Basic organisation present.	Code is difficult to read with poor naming conventions. Inconsistent or chaotic structure.
Modularity and Reusability (9%)	Excellent use of modular, reusable components. Components are well-abstracted and single-purpose. No code duplication. DRY principles followed throughout.	Good modular structure with reusable components. Minimal code duplication. Most code follows DRY principles.	Some modular components present. Some code duplication exists. Partial adherence to DRY principles.	Little modularity. Significant code duplication. Poor component abstraction.
Code Simplicity and Clarity (8%)	Code is simple, elegant and easy to understand. Complex logic is well-explained. Appropriate use of React Native features without over-engineering.	Code is generally simple and clear. Most logic is easy to follow with minor complex sections.	Code works but contains unnecessarily complex sections. Some logic is difficult to follow.	Code is overly complex or convoluted. Difficult to understand the implementation.
Responsive Design (5%)	Applications are fully responsive across different screen sizes and orientations. Layouts adapt seamlessly. Excellent use of flexbox and responsive units.	Applications mostly responsive with minor issues on some screen sizes or orientations. Good use of responsive design.	Basic responsive design present. Some screens adapt but with layout issues.	Poor or no responsive design. Layouts break on different screen sizes or orientations.
Performance and Resource Management (4%)	Excellent resource management. Efficient data handling and state management. Proper memory management. Optimised lists and images. Smooth animations and transitions.	Good resource management. Mostly efficient implementations. Adequate optimization with room for improvement. Minor performance issues.	Basic resource management. Some inefficient implementations. Performance acceptable but not optimised.	Poor resource management. Inefficient data handling. Performance issues present. Memory leaks or slow rendering.

Version Control - Learning Outcome 3 (5%)

Criteria	Excellent (A)	Good (B)	Satisfactory (C)	Not Yet Achieved (D)
Commit History and Messages (5%)	Regular, meaningful commits throughout development. Clear, descriptive commit messages. Logical progression showing development process. Clean commit history in project branch.	Regular commits with mostly clear messages. Commit history shows good development progression with minor gaps.	Commits present but irregular or clustered. Some commit messages unclear. Basic development history visible.	Few commits or poor timing. Unclear messages (e.g., "update", "fix"). No clear development progression.

Testing - Learning Outcome 3 (10%)

Criteria	Excellent (A)	Good (B)	Satisfactory (C)	Not Yet Achieved (D)
User Testing Process (4%)	Six users tested (three per version). Well-structured testing sessions with clear tasks. Comprehensive written feedback collected covering usability, UI and bugs. Professional documentation.	5-6 users tested. Testing sessions conducted with mostly clear tasks. Good written feedback collected covering most areas.	3-4 users tested. Basic testing conducted. Some written feedback collected but incomplete.	Fewer than 3 users tested or testing poorly documented. Minimal or no written feedback.
Testing Report Quality (6%)	Comprehensive report with all required sections. Tester information anonymised. Tasks clearly documented. Thorough analysis of positive feedback, improvement areas and bugs. Excellent suggestions for improvements with clear rationale.	Good report covering all required sections. Most elements well-documented. Adequate analysis of feedback and bugs. Good improvement suggestions.	Report present covering most sections. Basic documentation of testing. Some analysis present. Limited improvement suggestions.	Report missing, incomplete or significantly lacking detail. Poor documentation. No analysis or improvement suggestions.

Reflection - Learning Outcomes 1, 3 (5%)

Criteria	Excellent (A)	Good (B)	Satisfactory (C)	Not Yet Achieved (D)
Reflection Quality (5%)	Thoughtful reflection (300-500 words) covering all required points. Clear articulation of challenges, solutions and learning. Demonstrates deep understanding and self-awareness about mobile development process.	Good reflection covering all required points. Adequate discussion of challenges and learning with some depth. Within word count.	Reflection present covering most points. Brief or superficial discussion. May be slightly outside word count.	Reflection missing, incomplete or significantly outside word count. Lacks depth or doesn't cover required points.

Presentation - Learning Outcome 2 (5%)

Criteria	Excellent (A)	Good (B)	Satisfactory (C)	Not Yet Achieved (D)
Video Demonstration (5%)	Clear 5-10 minute recording demonstrating all v1 and v2 features. Well-paced, professional presentation. Easy to follow with good audio/visual quality. Shows app running on device or emulator. Link submitted correctly.	Good recording within time limit. Most features demonstrated. Generally clear with minor audio/visual issues. Link submitted.	Recording present but may be too short/long. Some features demonstrated. Quality acceptable but could be clearer.	Recording missing, significantly outside time limit, unclear, or doesn't demonstrate key features. Poor quality. Link missing.